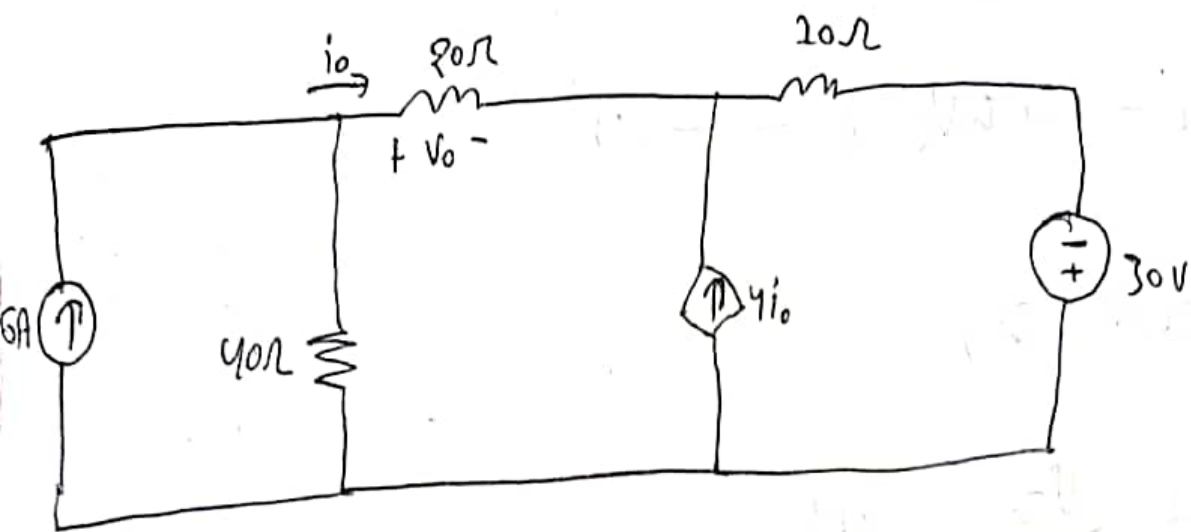


Oli Ahmed

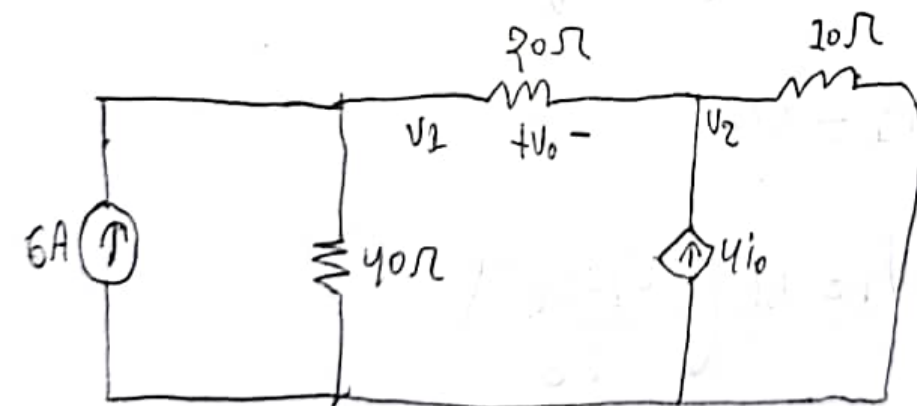
2017-3-60-073

AP 1

Ans to the ques No-2



When 6A active,



KCL apply at node 1,

$$\frac{V_1 - V_2}{20} + \frac{V_1}{40} = 6$$

$$\frac{2}{20} = 0.05$$

$$\frac{1}{40} = 0.025$$

$$\Rightarrow 0.05V_1 - 0.05V_2 + 0.025V_2 = 6$$

$$\Rightarrow 0.075V_1 - 0.05V_2 = 6 \quad \text{--- (1)}$$

KCL at node 2,

$$\frac{V_2 - V_1}{20} + \frac{V_2}{20} = 4i_0$$

$$\Rightarrow 0.05V_2 - 0.05V_1 + 0.1V_2 = 4i_0$$

$$\Rightarrow 0.15V_2 - 0.05V_1 = 4$$

$$\Rightarrow 0.15V_2 - 0.05V_1 = 4 \left(\frac{V_1 - V_2}{20} \right)$$

$$\Rightarrow 0.15V_2 - 0.05V_1 = 0.2V_1 - 0.2V_2$$

$$\Rightarrow 0.35V_2 - 0.25V_1 = 0 \quad \text{--- (2)}$$

Solving (1) and (2) we get,

$$V_1 = 152.72 \text{ V}$$

$$V_2 = 109.09 \text{ V}$$

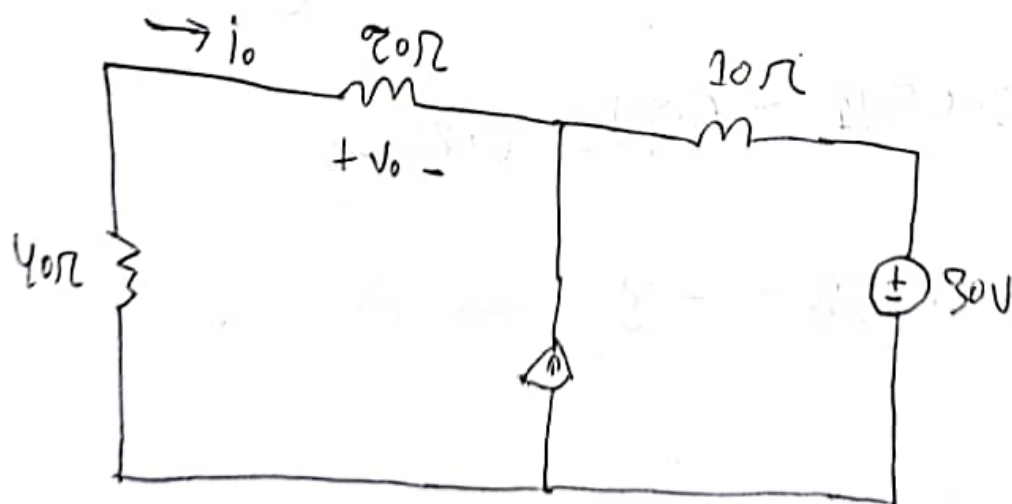
$$V_0' = V_1 - V_2$$

$$= 152.72 - 109.09$$

$$= 43.63 \text{ V}$$

Now,

When 30V active,



KCL at Node 1,

$$\frac{V_1}{40} + \frac{V_1 - V_2}{20} = 0$$

$$\Rightarrow 0.025 V_1 + 0.05 V_1 - 0.05 V_2 = 0$$

$$\Rightarrow 0.075 V_1 - 0.05 V_2 = 0 \quad \text{--- (i)}$$

KCL at Node 2,

$$\frac{V_1 - V}{20} + \frac{V_2 - 30}{20} = 4i_0$$

$$\Rightarrow 0.05 V_2 - 0.05 V_1 + 0.1 V_2 + 3 = 4 \left(\frac{V_1 - V_2}{20} \right)$$

$$\Rightarrow 0.25 V_2 - 0.05 V_1 = 0.2 V_1 - 0.2 V_2 - 3$$

$$\Rightarrow 0.25 V_1 + 0.35 V_2 = -3 \quad \text{--- (ii)}$$

Solving, (i) and (ii) we get,

$$V_1 = -20.9 \text{ V}$$

$$V_2 = -16.3 \text{ V}$$

$$\begin{aligned}V_0'' &= V_1 - V_2 \\&= -10.9 + 16.3 \\&= 5.4\text{V}\end{aligned}$$

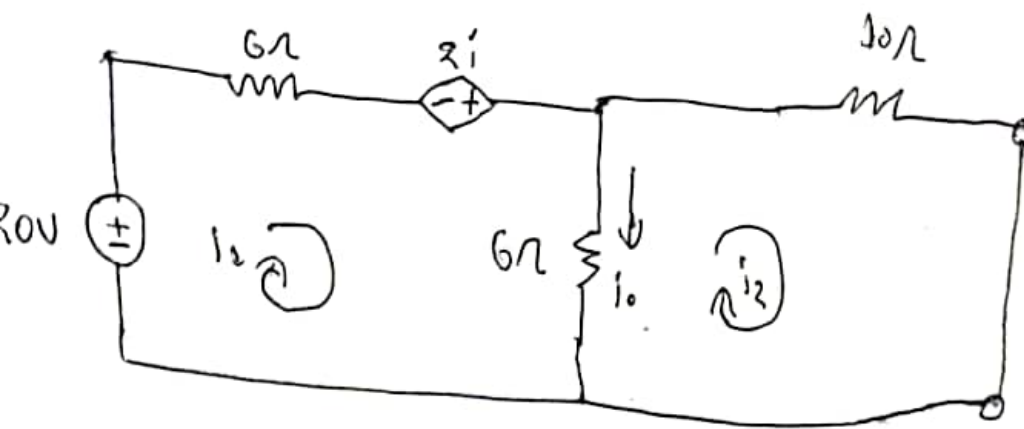
Using Super Position,

$$\begin{aligned}V_0 &= V_0' + V_0'' \\&= 43.63 + 5.4 \\&= 49.03\text{V}\end{aligned}$$

$$\begin{aligned}\therefore i_0 &= \frac{V_0}{R} \\&= \frac{49.03}{20} \\&= 2.45\text{A}\end{aligned}$$

Ans.

Ans to the Ques no-9



KVL at mesh 1,

$$6i_1 - 6i_2 + 6i_1 = 20 + 2i_1$$

$$\Rightarrow 12i_1 - 6i_2 - 2(i_1 - i_2) = 20 \quad [i' = i_1 - i_2]$$

$$\Rightarrow 12i_1 - 6i_2 - 2i_1 + 2i_2 = 20$$

$$\Rightarrow 10i_1 - 4i_2 = 20$$

$$\Rightarrow 5i_1 - 2i_2 = 10 \quad \text{--- (1)}$$

KVL at mesh 2,

$$6i_2 - 6i_1 + 10i_2 = 0$$

$$\Rightarrow 16i_2 - 6i_1 = 0$$

~~$$\therefore 3i_1 - 8i_2 = 0$$~~

~~$$3i_1 - 8i_2 = 0$$~~

$$\therefore 3i_1 - 8i_2 = 0 \quad \text{--- (1)}$$

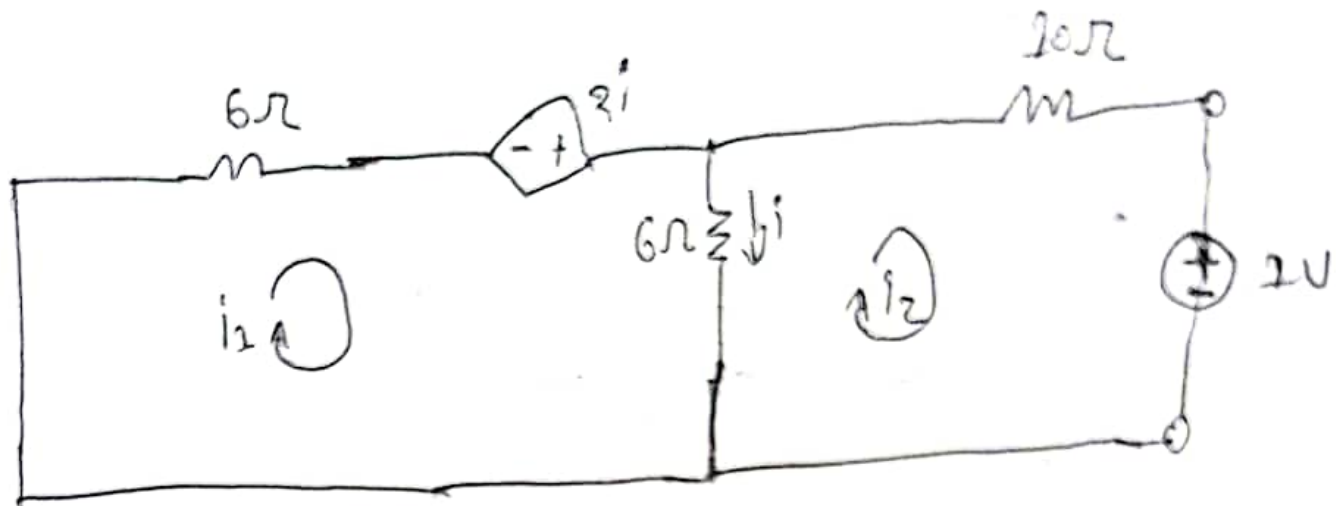
Now,

$$i_1 = \frac{40}{17}$$

$$i_2 = \frac{25}{17}$$

$$\therefore I_n = \text{Norton's Current} = i_2 = \frac{25}{17}$$

Now,



KVL at mesh 1,

$$6i_1 + 6i_1 - 6i_2 = 2i_1$$

~~$$\Rightarrow 6i_1 + 6i_1 - 6i_2 - 2i_1 = 0$$~~

$$\Rightarrow 6i_1 + 6i_1 - 6i_2 - 2(i_1 - i_2) = 0$$

$$\Rightarrow 6i_1 + 6i_1 - 6i_2 - 2i_1 + 2i_2 = 0$$

$$\Rightarrow 10i_1 - 4i_2 = 0$$

$$\Rightarrow 5i_1 - 2i_2 = 0 \quad \text{--- (1)}$$

KVL at mesh 2,

$$6i_2 - 6i_1 + 10i_2 = -1$$

$$\Rightarrow 6i_1 - 16i_2 = 1 \quad \text{--- (1)}$$

From (1) and (1),

$$i_1 = -\frac{1}{34} \text{ A}$$

$$i_2 = -\frac{5}{68} \text{ A}$$

for 20V,

$$\text{Current} = 20 \times \frac{5}{68}$$

$$= \frac{35}{17}$$

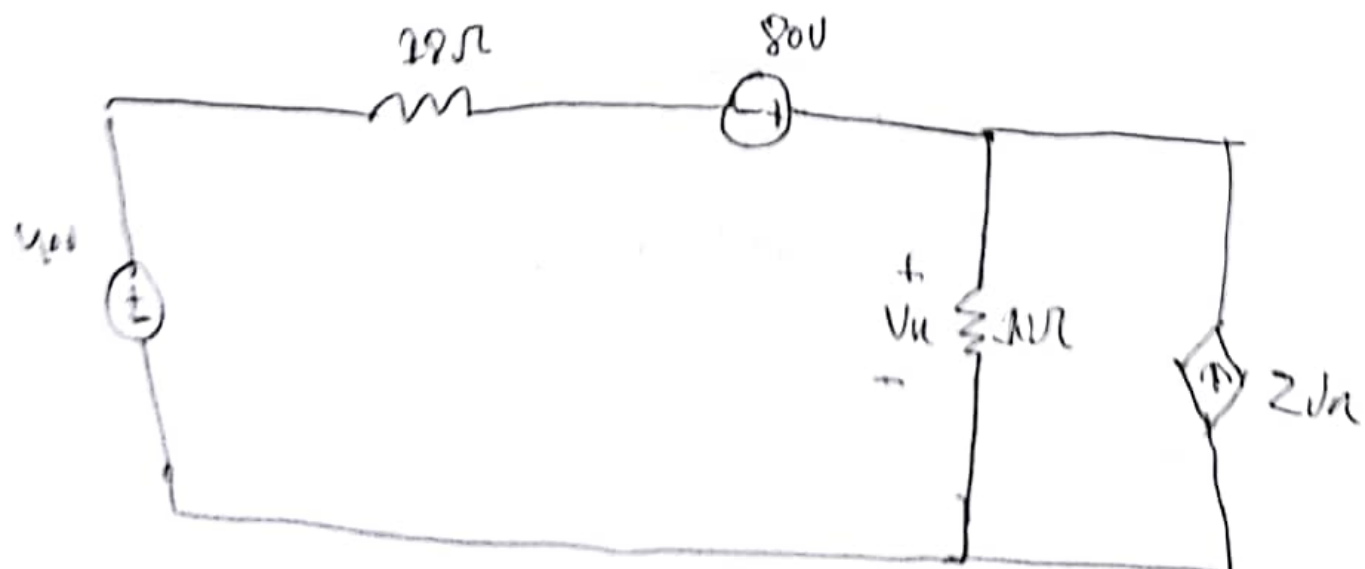
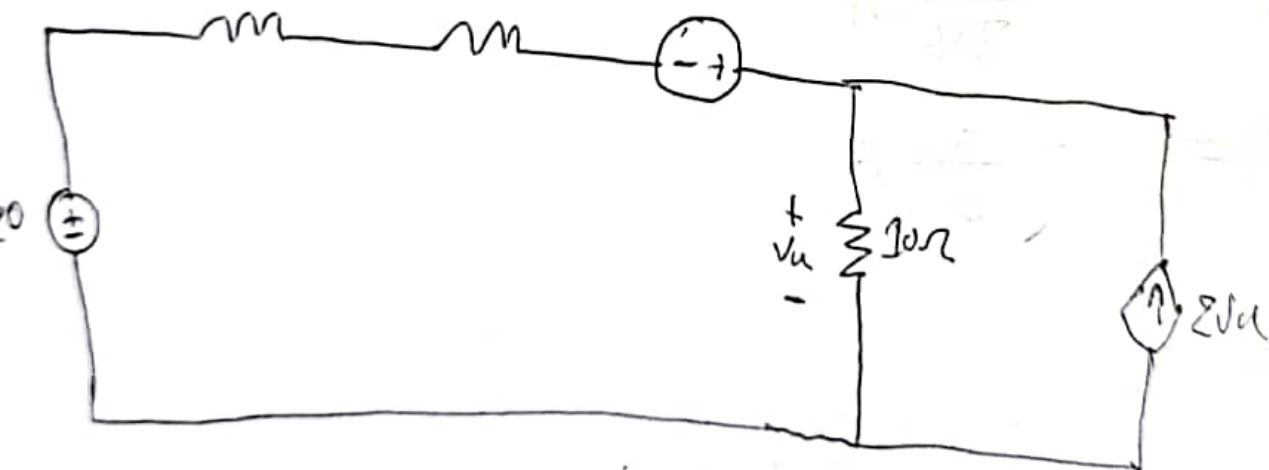
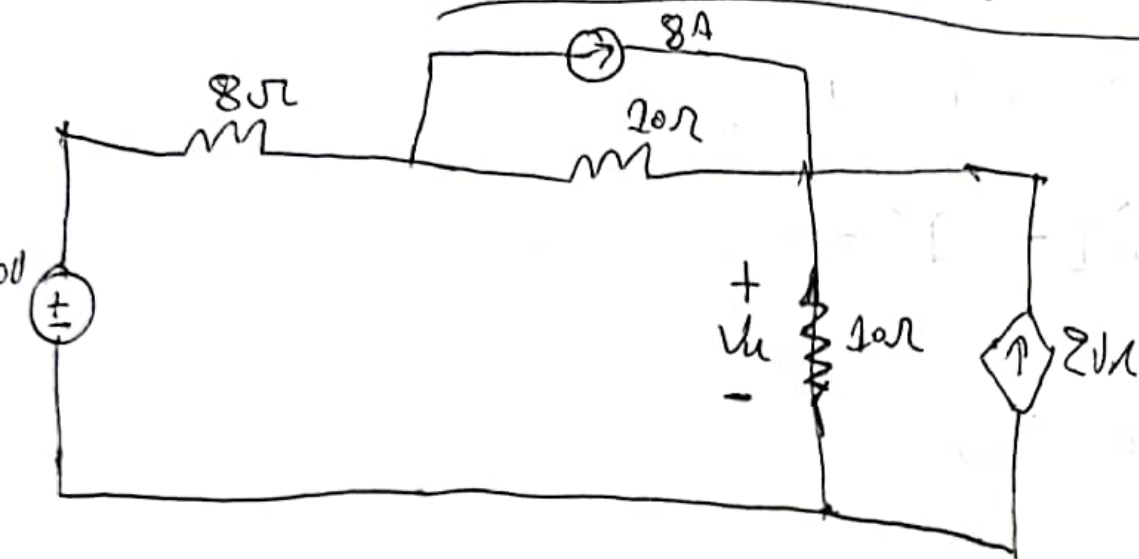
$$= 2.070 \text{ A}$$

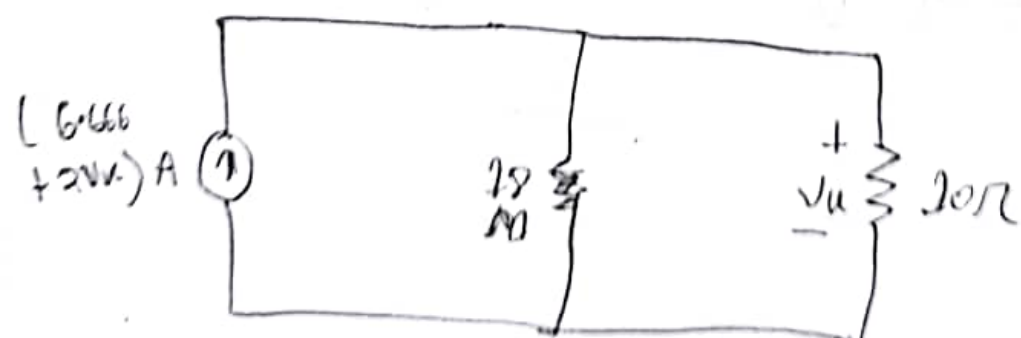
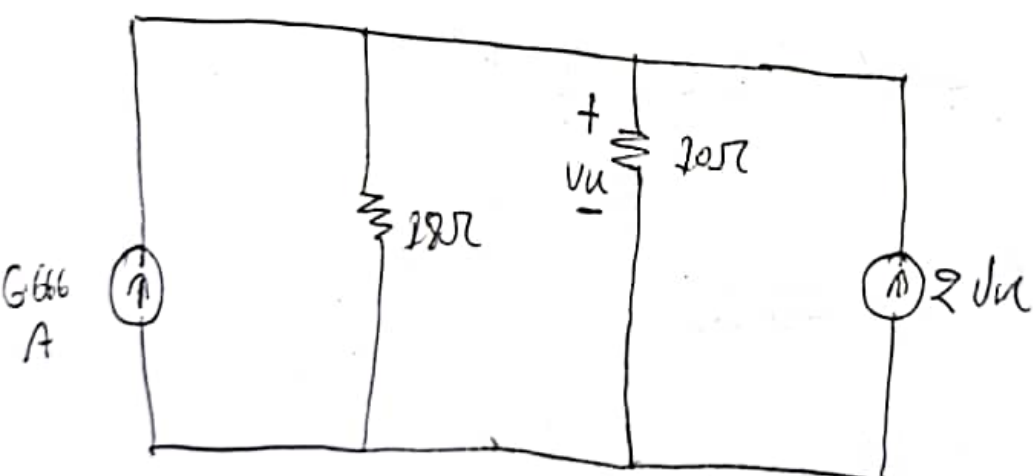
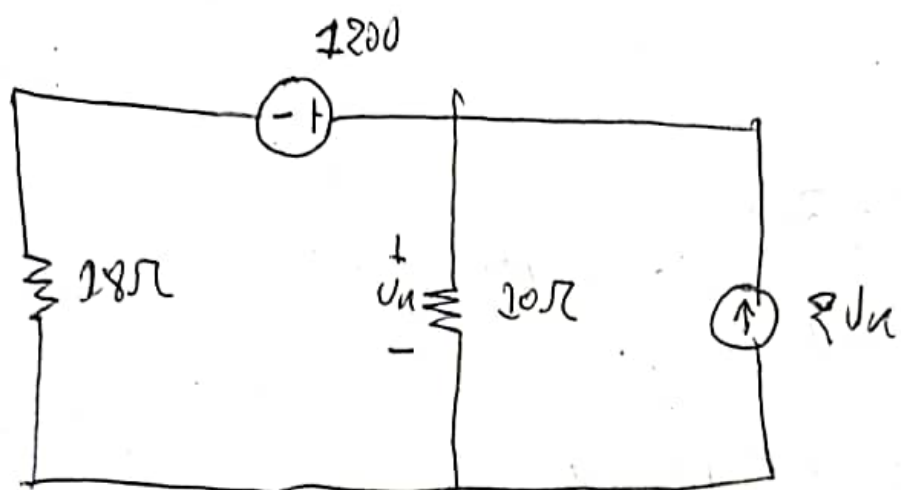
$$\therefore R_{th} = \frac{V}{i} = \frac{20}{2.070}$$

$$= 9.66 \Omega.$$

Ans.

Ans to the ques no-3





$$i_u = \frac{6.666 + 2v_u}{28 \times 10} \times 28$$

$$= 0.666 + 0.011 v_u$$

$$= 20 i_u$$

$$= 20 (0.666 + 0.011 v_u)$$

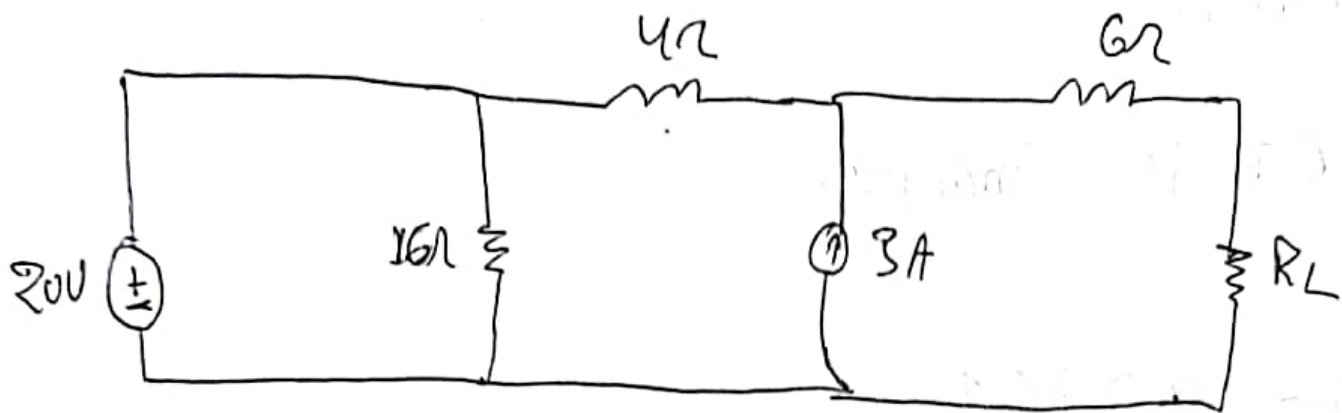
$$\therefore v_u = 6.66 + 0.22 v_u$$

$$\Rightarrow 0.78 v_u = 6.66$$

$$\Rightarrow v_u = 7.488 \text{ V}$$

Ans.

Ans to the ques no-5

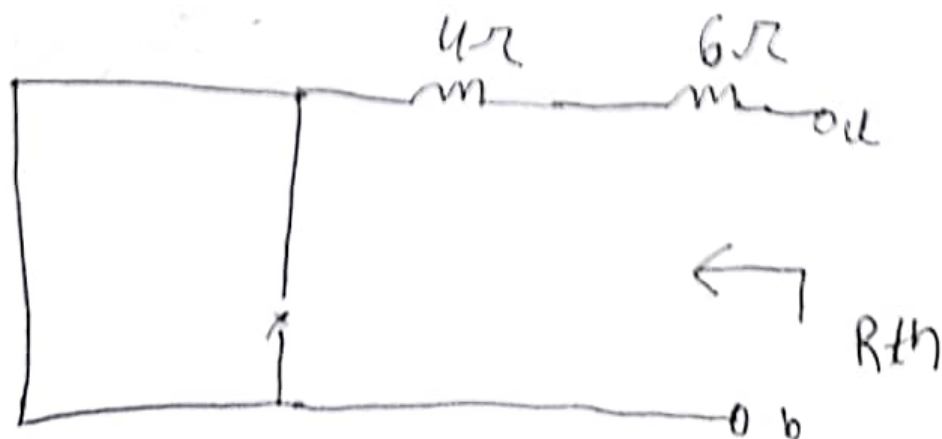


Nodal for V_{th} node -

$$\frac{V_{th} - 20}{4} = 3$$

$$\Rightarrow V_{th} = 32V$$

Find R_{th} -



6Ω resistance is parallel with short circuit.

$\therefore 6\Omega$ is inactive.

$$R_{th} = 4\Omega + 6\Omega \\ = 10\Omega$$

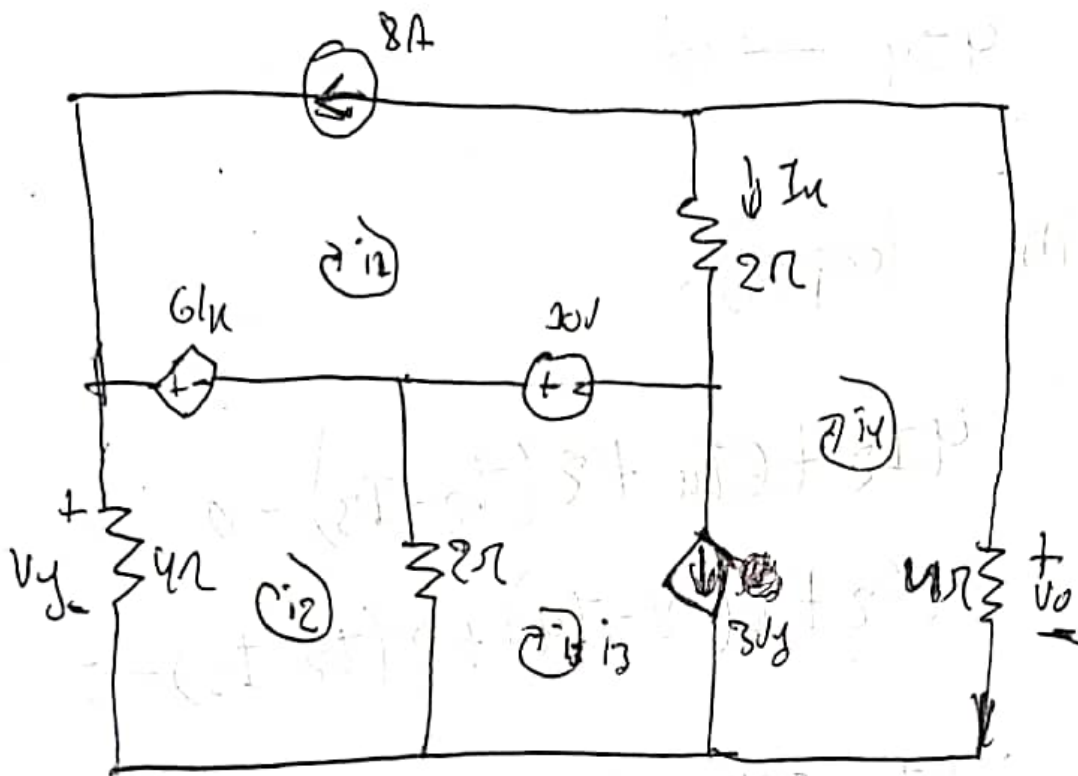
- short circuit not labored

Now,

$$\text{Maximum Power } P_{max} = \frac{V_{th}^2}{4 \times R_{th}} \\ = \frac{32^2}{4 \times 10} \\ = 25.6 \text{ W}$$

Ans-

And to the Ques No-2



from above circuit,

$$I_1 = -8A \rightarrow (1)$$

Now,

$$I_u = I_1 - I_4$$

$$I_u = -8 - I_4 \quad (11)$$

$$\therefore V_y = -4I_2 \text{ --- (11)}$$

$$V_o = 4I_4 \text{ --- (12)}$$

KVL in loop (2) $\rightarrow$

$$4I_2 + 6I_4 + 2(I_2 - I_3) = 0$$

$$4I_2 + 6(-8 - I_4) + 2(I_2 - I_3) = 0$$

$$\therefore 6I_2 - 2I_3 - 6I_4 = 48 \text{ --- (5)}$$

KVL in supermesh, loop (3) and (4) $\rightarrow$

$$2(I_3 - I_2) + 20 + 2(I_4 - I_2) + 4I_4 = 0$$

$$\Rightarrow 2(I_3 - I_2) + 20 + 2(I_4 + 8) + 4I_4 = 0$$

$$\therefore -2I_2 + 2I_3 + 6I_4 = -26 \text{ --- (6)}$$

at Super mesh,

$$I_3 - I_4 = 3V_y$$

$$I_3 - I_4 = 3(-4I_2)$$

$$\therefore 2I_2 + I_3 - I_4 = 0 \rightarrow (2)$$

By Solving

(5), (6), (7) we get,

$$I_2 = 5.5 A$$

$$I_3 = -52.375 A$$

$$I_4 = 14.625 A$$

$$\therefore V_o = 4I_4$$

$$= 4 \times (14.625)$$

$$= 58.5 V$$

Ans.